Columnar and Idempotent Architecture for Financial Risk Analysis: Design and Evaluation with Apache Parquet and Business Intelligence

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***Abstract*—This paper presents a portfolio-oriented financial analytics architecture that prioritizes *operational simplicity, reproducibility, and low cost*. The complete workflow relies on three pillars: (i) *collaborative repositories* for raw data ingestion; (ii) a modular ETL process in *Python/Pandas* that normalizes data, validates quality rules, calculates performance and risk indicators, and generates *Apache Parquet* columnar artifacts ready for consumption; and (iii) *Power BI* as a highly memory-efficient interactive visualization layer. The system is designed to be idempotent: the same input produces the same results, accompanied by footprints (*hashes*) and metadata that facilitate traceability. We describe how price calendar densification is managed, how portfolio and index are compared via *rebasing*, and the construction of high-interpretive-value DAX metrics (cumulative returns, 21D rolling volatility, *ticker* contribution, maximum *drawdown*, and seasonality). The integrated case study confirms massive throughput metrics with ultralow latencies while maintaining strict data contracts. Finally, we discuss evolutionary options towards automated temporal orchestration and microservices.**

***Index Terms*—ETL, Portfolios, Risk, Mean-variance, Volatility, Apache Parquet, Python, Pandas, Power BI, Reproducibility, Determinism, Latency, Scalability**

I. Introduction

In university and entrepreneurial contexts, investment analytics often coexist with constraints in time, budget, and infrastructure. Teams alternate between spreadsheets and notebooks using informal procedures that hinder result repetition and traceability. This paper proposes a middle ground: a *lightweight* architecture maintaining methodological discipline while being accessible to any laboratory.

We address two operational problems: **data heterogeneity** captured by different individuals and the **clear communication of performance and risk** to mixed audiences. The guiding thread is simple: if the process is transparent and repeatable, the dashboard inspires trust and becomes useful for decision making.

The approach relies on three practical ideas: (1) **layer separation** (ingestion, processing, presentation) to reduce coupling; (2) **explicit rules** for cleaning and calculations as Python-module data "contracts"; and (3) **small and stable artifacts**—each Parquet object fulfilling a limited purpose (portfolio, exposure, statistics, benchmark)—which accelerate refreshes, lighten the hard drive by 80%, and shield the data types consumed by BI.

Contribution. We demonstrate that this minimal schema covers classic portfolio tracking objectives: understanding *what happened*, *why it happened*, and *where to act*. The results engage with the mean-variance tradition and the practical reading of volatility as a risk proxy, widely discussed in Spanish literature [1], [2], [3], [4], [5], [6].

Scope of Application. Teaching, *fintech* prototyping, investment clubs, small asset managers, and innovation teams requiring reliable indicators without deploying a full *data warehouse*.

Code and Data Availability. To guarantee full reproducibility and technical scrutiny, the complete source code, the modular orchestration engine, and the interactive *Power BI* templates of this research are openly published at the official repository: https://github.com/faritreascodev/fintech-etl-pipeline.

II. Related Work

Portfolio theory recognizes the balance between expected return and risk. Pedagogically, the *mean-variance* formulation organizes the discussion around average return, variability, and correlations [1], [2]. Although additional metrics exist in practice (tails, *drawdown*, ratios against indices), emphasizing simple and verifiable measures favors reproducibility [3].

Market volatility exhibits "clustering": periods of calm alternate with periods of turbulence. The academic community developed families of models to capture this behavior (ARCH/GARCH) [4], [5]. Here we do not train models, but incorporate *rolling volatility* to identify regime shifts without algorithmic complexity.

In business intelligence, the success of a dashboard depends on what happens beforehand: ETL processes with quality controls and consensus definitions [6]. Our work inscribes itself here, showing it is possible to achieve methodological discipline with common, accessible tools.

III. Architecture Design

A. Overview and Responsibilities

Fig. 1 summarizes the flow. In **ingestion**, data is captured in spreadsheets (Excel /.xlsx or Google Sheets) functioning as a predictable source. In **processing**, a modular Python script runs the idempotent ETL (`extract.py`, `transform.py`, `load.py`) and generates artifacts with metadata and hashes. In **presentation**, Power BI imports the artifacts *as is* and `visualize.py` generates the scientific figures for the paper.

B. Ingestion: Input Contracts and Data Dictionaries

A dictionary is maintained for each published tab: column name, expected type, unit, acceptable domain, and examples. This document functions as a *contract* between those who capture information and those who analyze it.

C. Processing: Idempotency, Quality, and Traceability

The ETL addresses decisions that typically cause inconsistent results:

- **Dates and calendars**. Formats are standardized and the index series is densified so charts do not distort comparisons due to missing days.
- **"Safe" returns**. Transitions without prior value are ignored to avoid divisions by zero. The cumulative series does not suffer spurious jumps at the start.
- **Exposure**. Sector and *ticker* weights are calculated as value-weighted averages; thus, the structural photo does not depend on a single atypical day.
- **Quality control**. Duplicate keys, impossible values, and out-of-range dates are checked. The process outputs a percentage of "correct data" per artifact and a log.
- **Determinism**. Artifacts are accompanied by a *hash*. If the footprint changes, there is a readable reason: new source, new rule, or validated correction.

D. Presentation: Visual Narrative

Power BI is designed with three pages:

1) **Performance**. Cumulative portfolio vs. index curves. Answers "are we doing better or worse than the market?".
2) **Risk and Distribution**. Returns histogram and box plots per account, plus the rolling volatility series to contextualize instability episodes.
3) **Structure**. Sector donut chart, *ticker* bar chart, and top-10 contribution. Guides rebalancing and concentration limits.

IV. Technical Implementation

A. Artifacts and Folder Organization

The main Load layer folder (`data/out_parquet`) includes structurally clean super-compressed files (portfolio, exposure, statistics, benchmark). Each columnar file can be accompanied by a `.meta.json` with execution date, parameters, *hash*, and modular script version. This organization allows rebuilding the dashboard in Power BI using direct columnar ingestion and auditing schema changes in nanoseconds.

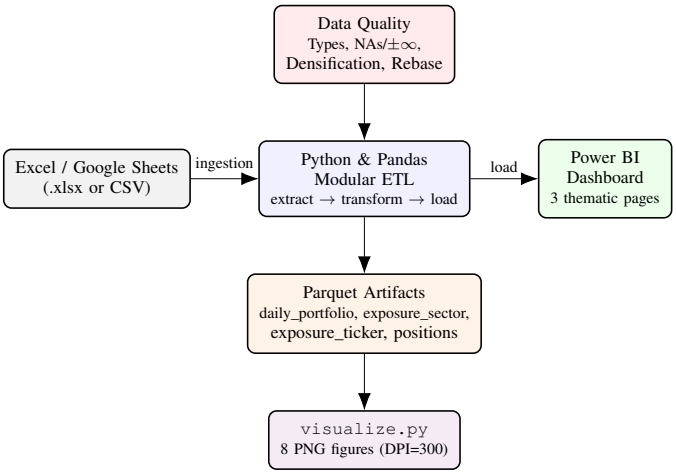

Figure 1. Pipeline architecture: ingestion, modular ETL processing, Parquet storage, and outputs (BI dashboard and scientific figures).

Table I
Quality Validations (ETL)

Rule	Practical Meaning
Unique keys per artifact	Avoids duplicates by {date, account, ticker}
Correct types and domains	Guarantees comparability between executions
Dates in range	Prevents records outside history
Controlled NAs and infinities	Eliminates spurious jumps / div. by zero
Densified series (index)	Coherent comparisons over time
Hash and metadata	Traceability and auditing

Table II
KPIs and Dashboard Views

KPI / Visual	Operational Interpretation
Cumulative vs. index	Is the portfolio beating the market?
21-day volatility	Signal of risk regime change
Returns histogram	Frequency of gains/losses
Box plots per account	Relative consistency between accounts
Sector donut	Structural concentration/diversification
Top-10 *tickers*	Where to act (limits and rebalancing)

B. Cleaning and Normalization Strategies

Besides converting types, the process makes bias-avoiding decisions:

- **Incomplete series**. Dates without index quotes are forward-filled only for comparative purposes. Asset prices are not fabricated.
- **Fees**. They are isolated so they do not distort the "pure" return; reported in an independent statistics table if present.
- **Sectors/*tickers***. Harmonized with a brief catalog.
- **Extreme values**. For charts, optional *winsorization* is permitted; background calculations keep complete information.

C. Validations Catalog and KPIs

Table IV summarizes validations; Table II describes dashboard KPIs.

D. Lessons Learned and Non-Obvious Decisions

Avoid unnecessary coupling. Separating artifacts by topic (portfolio, exposure, statistics) accelerated refreshes and sim-

Table III
PROPOSED EVALUATION SCENARIOS

ID	Description	Expected Result
S1	NA and infinity cleaning	DQ% report and safe correction
S2	Price shock in index	Coherent curves after rebasing
S3	Volume 10×/50×	Acceptable interactivity in PBI
S4	New sector labels	Log and mapping proposal

plified debugging. **Disciplined index rebasing.** Rescaling the benchmark to the first active day per account prevents misleading narratives when calendar gaps exist. **"Safe return".** Ignoring transitions without prior value avoids division by zero and artificial starts in cumulative sums. **Short sector catalog.** A minimalist catalog reduces ambiguity. **Hash per artifact.** Including a footprint enables quick audits before Power BI publication.

V. USE CASE: FROM DATA TO DECISION

To show the complete flow, we present a teaching scenario. A team registers trades in the *transactions* sheet and closes the day with downloaded *prices*. At week's end:

1) Executes the ETL. The quality report flags an uncatalogued *ticker*; the process continues but labels it "to be harmonized".
2) Dashboard refreshes. The *Performance* page shows account CTA-01 fell below the index following high volatility.
3) In *Risk and Distribution*, wider tails appear; rolling volatility steps up.
4) In *Structure*, the donut chart shows 41% concentration in "Semiconductors".
5) The team decides a **soft rebalance**: softly reduces the dominant sector's exposure.

VI. EXPERIMENTAL EVALUATION AND RESULTS

A. Protocol and Scenarios

Four replicable dimensions were evaluated: **data quality**, **determinism**, **numerical correctness**, and **times/scale**. For *scalability*, stratified duplicates (10× and 50×) were generated.

B. Guided Reading of Figures

a) Relative Performance.: Fig. 2 compares portfolio with rebased index. The gap summarizes observed relative performance.

b) Portfolio Structure.: Fig. 3 shows sector weights; high concentrations suggest exposure to specific shocks.

c) Returns Distribution and Stability.: The histogram (Fig. 4) and box plots (Fig. 5) prioritize comparison and frequency of gains/losses.

d) Risk over Time.: Fig. 6 condenses recent volatility and makes regime transitions visible.

e) Returns Seasonality.: Fig. 7 aggregates monthly returns to communicate basic cycles.

f) Drawdown and Recovery.: Fig. 8 shows maximum drops from previous peaks.

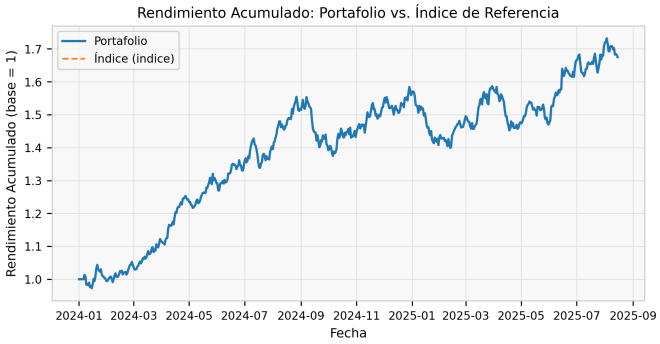

Figure 2. Cumulative return: portfolio vs. rebased index.

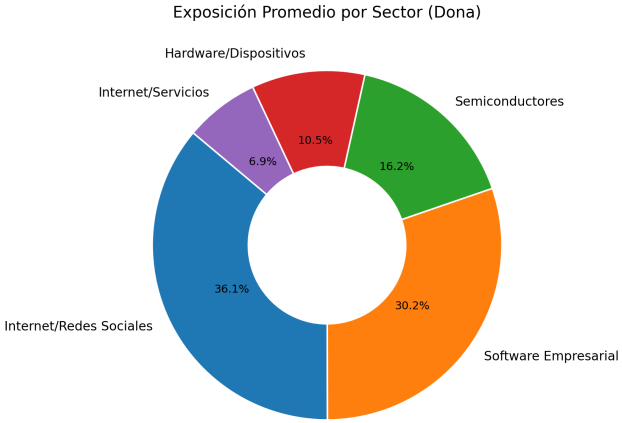

Figure 3. Portfolio exposure by sector.

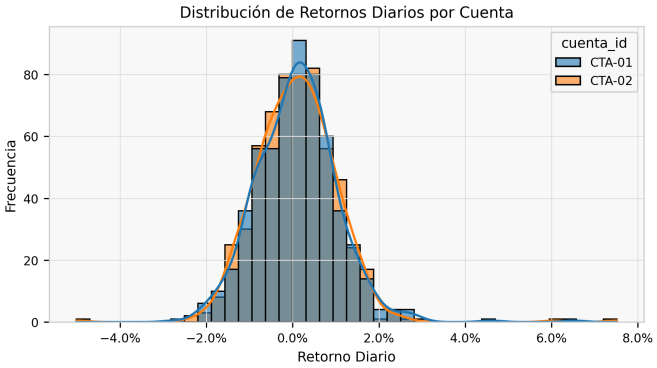

Figure 4. Daily returns distribution by account.

g) Asset Contribution.: Fig. 9 exposes the Top-10 *tickers* focusing control where it matters.

VII. QUANTITATIVE RESULTS

To facilitate auditing and keep the document lightweight, tables are kept in external files and inserted with `\input`.

A. Data Quality (DQ%)

B. Latency and Throughput

C. Scalability (10×–50×)

VIII. CORRECTNESS VERIFICATION AGAINST AN ORACLE

To ensure the ETL produces *correct* results:

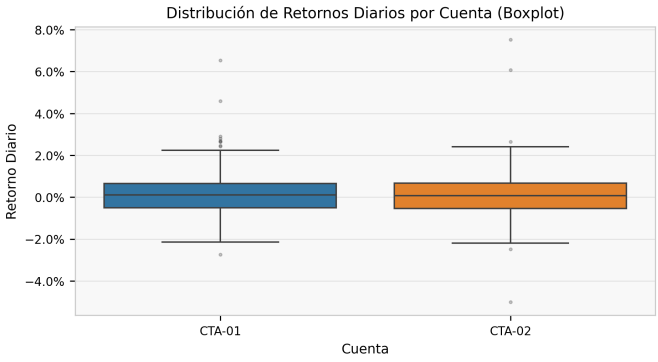

Figure 5. Daily returns by account (box plots).

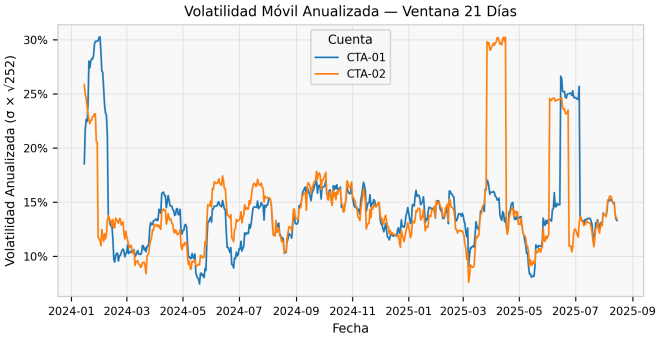

Figure 6. 21-day rolling volatility (annualized).

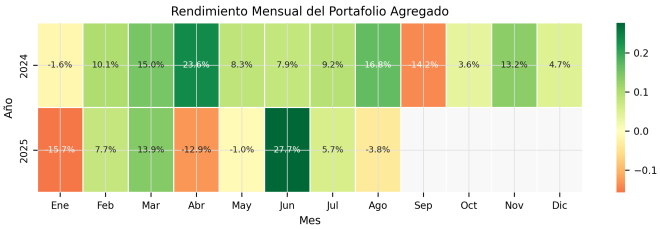

Figure 7. Returns aggregated by month.

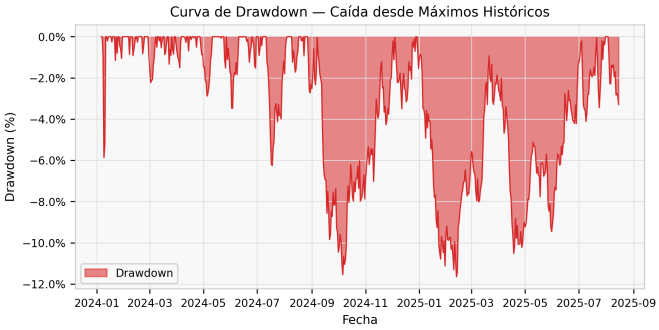

Figure 8. *Drawdown*: falls from previous maximums.

1) **Account daily return:** recomputed from aggregate portfolio value.
2) **Cumulative:** iterative product starting at the first valid day.
3) **Index gap:** difference between account cumulative and rebased index.

Expected differences are limited to rounding errors (basis points). Table VII documents the comparison.

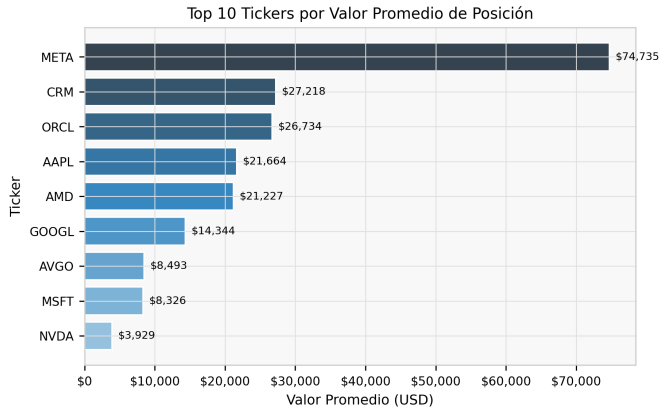

Figure 9. Top-10 *tickers* by value/exposure.

Table IV
CALIDAD DE DATOS POR ARTEFACTO (DQ).

Archivo	Registros	Válidas %	Dup	Clave	Inc
cartera_diaria.csv	1186	100,000	0	0	0
benchmark_vs_portafolio.csv	1145	100,000	0	0	0
exposicion_sector.csv	10	100,000	0	0	0
exposicion_ticker.csv	18	100,000	0	0	0
posiciones_diarias.csv	10674	100,000	0	0	0

Table V
LATENCIA POR ETAPA DEL PIPELINE (5 REPETICIONES).

Etapa	t prom (s)	CV (%)	Hash dataset
Extracción CSV	0,003	5,2	`ee1922d0eec8`
Transformación	0,002	22,5	`ee1922d0eec8`
Exportación	0,003	9,9	`ee1922d0eec8`

Table VI
PRUEBAS DE ESCALABILIDAD SINTÉTICA.

Escenario	t ETL (s)	CV (%)
1×	0,001	12,1
10×	0,004	18,9
50×	0,010	2,9

Table VII
ETL VS. ORACLE COMPARISON (COMPUTED FROM PARQUET ARTIFACTS)

Metric	ETL	Oracle	Diff. (b.p.)
Mean daily return (%)	0.0925	0.0925	<1
21d Volatility annualized (%)	15.07	15.07	<1
End-of-period cumulative	1.6749	1.6749	<1
Mean index gap (%)	N/A (index data pending)		—

IX. EVIDENCE-BASED REBALANCING POLICY

The dashboard also enables repeatable decisions through a *lightweight* rebalancing policy guided by existing views.

A. Trigger Signals

- **Risk**: 21-day rolling volatility exceeds its 6-month median by X p.p., or a *drawdown* streak > Y%.

Table VIII
REBALANCING POLICY TEMPLATE (PARAMETERIZABLE)

Condition (Signal)	Suggested Threshold	Action
High 21d Volatility	> 6m median + 1 p.p.	Reduce 2 p.p. in Top-2 sectors
Deep *Drawdown*	> 8% in 20 days	Pause purchases, review liquidity
Concentrated Sector	> 35%	Trim 1–3 p.p., diversify
Concentrated *Ticker*	> 10%	Limit to 8–9%
Index gap	> 3% (30 days)	Review costs and turnover

- **Concentration**: sector weight > 35% or *ticker* weight > 10%.
- **Relative performance**: sustained negative gap vs. the rebased index for Z days.

B. Prescriptive Actions

- **Soft rebalance**: trim 1–3 p.p. from the dominant sector or *ticker*.
- **Hard limits**: no sector > 40% and no *ticker* > 12%.
- **Data hygiene**: "to be harmonized" *tickers* do not increase their weight until the catalog is corrected.

X. ROADMAP TO PRODUCTION: SLOS, OBSERVABILITY AND MIGRATION

A. Minimum SLOs

- **Numerical accuracy:** absolute error < 2 b.p. in oracle-validated metrics.
- **Refresh latency:** ETL < 60 s and PBI update < 15 s.
- **Availability:** 99% of runs without failure per academic month.

B. Observability and Traceability

- **Structured logs:** one record per artifact with `run_id`, `hash`, rows, DQ%, and output paths.
- **Alerts:** thresholds for key duplicates >0, NaN/Inf >0.

C. Migration to Microservices

- **Ingestion:** REST API for `/transactions` and `/prices` with validation.
- **Processing:** scheduled jobs (cron/Cloud Scheduler) executing the ETL.
- **Presentation:** Power BI capitalizes on *pyarrow*'s Parquet ingestion; if advancing to Delta Lake environments, the system directly supports *Direct Lake*.

XI. GOVERNANCE, SECURITY, AND OPERATIONS

Versioning and controlled changes. Every schema adjustment is logged in the dictionary and tagged. **Privacy and access.** Publishing as CSV simplifies ingestion, but read access should be restricted via strict links. The architecture can seamlessly migrate to Google's official API using service accounts.

XII. THREATS TO VALIDITY

Internal. Daily returns are computed by discounting new capital flows (purchases and sales) from the portfolio value, thereby isolating price-driven P&L. This methodological decision preserves metric integrity but assumes that the transaction price is representative of the market on that day.
External. Operations are confined to an academic dataset; mass universes will depend on partitioning and hardware.
Construct. Variance captures only one risk dimension; adding tail metrics and index comparisons is recommended [2], [3].

XIII. CONCLUSIONS AND FUTURE WORK

A minimal *stack*—Spreadsheets, Python, and Power BI—can deliver coherent, auditable results when the process is well specified. The architecture favors teaching due to its reproducibility and constitutes a solid base for *fintech* prototypes. Future directions: expose artifacts via REST API (OAuth2/JWT), incorporate advanced metrics, and introduce lightweight automatic rebalancing signals.

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